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Complaint and Severity Identification from Online Financial Content

Apoorva Singh, Rohan Bhatia, and Sriparna Saha

***Abstract*—The automatic detection of financial complaints can benefit businesses and online merchants. Compared to manually tagged complaints, they can use this information to monitor and address issues and effectively route them to appropriate teams. This can also promote greater transparency and accountability when dealing with consumer financial products and services, strengthening the firm's brand value. In linguistic studies, complaints have been classified into severity categories based on the level of risk the complainant is prepared to accept. Furthermore, since emotions influence every speech act, an individual's emotional state considerably impacts the complaint expression. In this paper, we introduce a Financial Complaints resource, a collection of annotated complaints arising between financial institutions and consumers expressed in English on Twitter. The dataset has been enriched with the associated emotion, sentiment, and complaint severity classes. The dataset comprises 3149 complaint and 3133 non-complaint instances spanning over ten domains (e.g., credit cards, mortgages, etc.). For a comprehensive evaluation of our dataset, we develop a multi-task framework for complaint detection and severity classification with emotion recognition and sentiment classification as the additional tasks and compare it with several existing baselines[1].**

***Index Terms*—Complaint Corpus, Complaint Detection, Complaint Severity Detection, Deep learning, Finance Complaints, Multi-task learning**

I. Introduction

ARTIFICIAL INTELLIGENCE (AI) is transforming how people and organizations access and manage their finances. While the shift from traditional financial services to digital finance was already underway pre-pandemic, the pandemic accelerated the process as stay-at-home requests became the new normal for financial institutions, and customers decided to seek more self-service options. AI and machine learning in finance cover everything from fraud detection to chatbot assistants and task automation. Other than these mainstream applications, addressing digital service complaints could be a significant commercial application for AI in finance. Compared to manually tagged complaints, automated detection of financial complaints pertaining to fraudulent transactions, delayed fund transfers, lousy customer service, etc., could help direct them to appropriate teams. This can further help promote fairness and transparency while dealing with loans, credit cards, and other consumer financial products and services, thus improving customer experience and increasing the organization's brand value.

In linguistics, complaining is defined as an individual's statement of dissatisfaction with an enterprise, commodity, or event [1]. Complaints have been grouped into various degrees of severity based on the emotional intensity of the complaint, the amount of face-threat that the complainer is willing to undertake, and the complaint's motive [2], [3]. The objective of complaining could be to voice dissatisfaction, seek explanations, or both. Identifying complaints and associated severity levels in natural language is crucial for downstream application developers such as customer service chatbots [4] and commercial organizations to improve their customer support capabilities by identifying and resolving complaints [5].

Various social media and microblogging services, such as Twitter[2], Facebook[3], and Tumblr[4], have skyrocketed in popularity in recent years. The reason being all these forums enable effective communication amongst Web users worldwide. These social media sites, which offer millions of messages and information to be exchanged daily, are multifaceted and invaluable resources for investigating and understanding people's perceptions and evaluating the data generated by participants. Among various social media platforms, Twitter is among the most widely used social media services, with engaging individuals, famous events, and intriguing issues. The social aspect of Twitter, combined with its huge volume, has transformed it into a cost-effective data repository for exploring and evaluating activities, conduct, and user opinions. As a result, the crux of this study is analyzing Twitter-based complaints essentially focused on the Financial domain.

The intended content of every speech act is strongly influenced by a user's emotional condition [6]. Emotion detection is far more sophisticated and fine-grained in terms of research than sentiment classification [7]. Emotion provides a deeper understanding of the customer's mind when combined with sentiment. A statement, for example, could entail only a negative sentiment, whereas emotion could be anger or even disgust. The relationship between

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[1]The corpus and code will be made publicly available in the camera-ready version.

[2]http://twitter.com/

[3]https://facebook.com/

[4]https://tumblr.com/

emotion and sentiment urges us to examine the sentiment and emotion of customers while studying complaints.

The following are the major contributions of our proposed work:

- We curate a new corpus *Financial Complaints* (*FINCORP*) for aiding complaint identification and complaint severity classification research.
- We perform good quality annotations for the *FINCORP* dataset for solving four tasks: complaint detection, severity level detection, emotion recognition, and sentiment analysis.
- We present a framework for joint learning of (a) binary complaint classification, (b) complaint severity level classification, (c) emotion recognition, and (d) sentiment analysis. Complaint Identification (CI) and Complaint Severity level classification (CS) are the primary tasks in our multi-task framework. In contrast, Emotion Recognition (ER) and Sentiment Analysis (SA) are considered additional tasks. According to experimental results, the proposed approach surpasses the baselines and state-of-the-art approaches.

The rest of the paper is structured as follows. We present a brief overview of the existing literature in Section 2. In Section 3, we go over the specifics of the proposed dataset. In Section 4, we describe our proposed technique for multi-task experiments. The experimental setting and results analysis are covered in Section 5. Finally, we conclude in Section 6 with a discussion on future work prospects.

II. Related Work

This section discusses some of the prior works on the complaint identification task and the publicly available annotated complaint corpora.

A. Complaints in Linguistic Studies

Previous work in linguistics has categorized complaints based on their severity and directness. Complaints can be divided into four fine-grained severity classifications, according to Trosberg [2]: (a) no explicit reproach; (b) disapproval; (c) accusation; and (d) blame. Quite recently, [3] divided complaints into three categories: very direct, moderately direct, and indirect. Clear violations of expectations are referred to as direct complaints (i.e., very direct and moderately direct). On the other hand, indirect complaints do not directly specify or imply a breach of expectations. Furthermore, the contrast between very direct and moderately direct is that the former underlines the obligation of the complaint recipient, while the latter does not.

B. Multi-task Learning

Previous findings [8], [9], [10] have confirmed the efficiency of multi-task systems by simultaneously learning multiple associated tasks. An individual's emotional state and sentiment have a decisive impact on the intended content [6]. Along with sentiment, emotion offers a deeper understanding of the consumer's mindset [11], [8]. The correlation between emotion and sentiment motivates us to consider complainants' sentiment and emotion while analyzing complaints. Moreover, sentiment and emotion play significant roles in human interactions and thereby contribute toward building efficient and versatile artificial intelligence-based systems [7].

Table I
Statistics of the existing and proposed dataset for complaint identification

Dataset	Size	Description
Complaints [12]	3449 instances out of which 1235 are complaints	Sample instances collected from Twitter grouped in nine generic domains.
Product Review [13]	3449 instances out of which 566 are complaints	Product Reviews collected from Amazon and Youtube websites.
FINCORP	6282 instances out of which 3149 are complaints	Sample instances collected from Twitter based on financial product type & concerns.

C. Complaint Identification

Complaint identification on social media is a time-consuming and challenging process that needs identifying complaints from disparate and noisy text samples with character constraints, the use of unpredictable abbreviations, and sarcastic expressions. Text-based complaints have earlier been investigated via semi-supervised strategies, complex feature engineering-based machine learning methods, and deep learning models [14], [12], [15]. In the work [12], authors proposed a logistic regression model with hand-crafted features for detecting complaints. The authors of the work [16] used an ensemble of neural language models enhanced by the usage of transformer networks on a Twitter-based Complaint dataset. In linguistics, complaints were earlier classified based on responsible branch, the level of urgency and the possibility of receiving a response [17], [18], [19].

D. Complaint Datasets

The *Complaints* dataset[5] was published in the work [12]. The authors selected 93 customer service handles from Twitter and sampled original tweets addressed to these accounts. Only tweets that received a response from the customer support accounts were considered. A total of 1971 tweets were collected using this method. Additionally, 1478 tweets were sampled to guarantee a diverse and representative range of tweets. Overall, the

[5]https://github.com/danielpreotiuc/complaints-social-media

Table II
ANNOTATION EXAMPLES FOR EACH COMPLAINT SEVERITY LEVELS

Severity level	Annotation Example
Accusation	I ordered a mattress from your site. Item is missing now. You are doing fraud.
Blame	One payment was made with my consent the another payment seems a fraud transaction gone to an unkown account.
Disapproval	Can't even begin to tell how disappointed I am with crypto market. Lost so much money.
No explicit reproach	I have a problem and need help, support site kept putting me in a loop. Help!

Table III
ANNOTATION EXAMPLES FOR EACH EMOTION CLASS

Emotion	Annotation Example
Anger	Most of the time your ATM machines does not work. Over that you will charge ATM usage charges also! What the hell is your problem?
Disgust	What kind of service is this? Absolutely pathetic and disgusting packaging. I want my money back.
Fear	I am scared due to this pandemic and lockdown will worsen our world economy.
Happiness	I love your mobile app, makes it so easy to do all my payments from one place. Awesome!
Sadness	I am not able to use netbanking and mobile since last evening. When will downtime end?
Surprise	Best investment I've ever made. The knowledge and understanding there is amazing!!
Other	Can we apply debit card in our nearest branch or only in home branch and what documents are required for this purpose?

dataset consists of 2214 non-complaints and 1235 complaints in English. The dataset was grouped into nine domains (i.e., Apparel, Cars, Electronics, Food, Retail, Service, Software, Transport, and Other). Recently in the work [15], authors extended the *Complaints* dataset with the complaint severity classes and analyzed the effect of severity levels on CI.

The *Product Review* corpus[6] [13] is a Hindi language-based product review corpus. This corpus is a collection of product reviews posted on the retail website Amazon[7] and the video-sharing platform, YouTube[8] comment section. The corpus comprises 3711 instances, with 3145 being non-complaints and 566 being complaints. The dataset was further grouped into five domains (i.e., Book, Headphones, Phone, Watch, and Miscellaneous).

[6]https://github.com/MrRaghav/Complaints-mining-from-Hindi-product-reviews/blob/master/amazon-youtube-hindi-complaints-data.xlsx
[7]https://www.amazon.in/
[8]https://www.YouTube.com/

Figure 1. One hundred most frequent words used for expressing complaints by Twitter users.

In this work, we introduce the *Financial Complaints* dataset (*FINCORP*), which unlike the *Complaints* and *Product Review* datasets, is a balanced dataset with nearly equal percentage of complaints (50.13%) and non-complaints (49.87%) samples. Unlike the other two datasets, which contain generic complaints from various domains, our dataset focuses on grievances arising between financial institutions and consumers on the social media network Twitter. Another unique feature of our dataset is that, in addition to the complaint identification axis, we have expanded the dataset on three other closely related axes (severity level classification, sentiment classification, and emotion recognition) to better understand the nuances of the complaint identification task. To the best of our knowledge, this is the first gold standard complaint dataset specifically dealing with Financial complaints.
We create a word cloud to graphically depict the frequency of words that appear more frequently in complaints expressed on Twitter. The more frequently a term occurred in the complaint tweets, the larger the term is in the image. Figure 1 depicts the top 100 terms used by 6282 Twitter users to convey their dissatisfaction. If we observe Figure 1 closely, we will see that Twitter users complain more frequently using key phrases like response, refund, and please, alongside other words like RBI, fraud, and help. Table I shows the details of various existing and proposed CI-related datasets.

III. CORPUS DESCRIPTION

The existing complaint datasets [12], [13] constitute only binary-class complaints. To encourage complaint identification and severity prediction research in the financial domain, we curate a *Financial Complaints* dataset (*FINCORP*) labeled with complaint labels, severity classes,

emotion, and sentiment classes. We chose Twitter as the data source as it is a forum that allows for a lot of self-expression, and users can communicate directly with each other or with corporate support handles. Tweets are freely available and are a popular choice for data analysis in various other similar tasks such as sentiment and emotion analysis ([9], [8]), etc. Here, we discuss the details of the FINCORP dataset.

A. Data Collection

Complaints account for a small percentage of all Twitter posts. Many companies employ Twitter to offer customer support and deal with customer complaints. To collect finance-related complaints and opinions from tweets, we use Tweepy[9] library to fetch tweets containing specific keywords and phrases using its *cursor class*. Based on an analysis of previous complaint datasets, we used a combination of keywords such as credit card/ debit card/ net banking/ UPI in combination with transaction failure/ fraud/ no response/ customer support/ complaint, etc., to fetch finance-related complaint tweets. Twitter Developers'[10] account is required to use this library, in which API tokens are provided by Twitter necessary to make the tweepy API object. In order to fetch tweets, queries in the form of strings have to be passed to the cursor object, and tweets containing those strings can be fetched and saved into data structures. We collected 6945 tweets to be considered for annotation using this technique.

Furthermore, we manually sorted the tweets into ten domains (Credit card, Customer service, Debit card, Economy, Financial policies, Investments, Loans, Retail banking, Salary, and Transaction failure) based on the financial product type and concerns expressed. We have done this to enable complaint analysis on a more granular level. In related studies on complaint analysis, tweets and product reviews were similarly sorted based on domains ([12], [13]).

We anonymize all usernames and URLs in the tweet and substitute them using placeholder tokens for preprocessing. Using langid.py ([20]), tweets were filtered for English, and retweets were discarded. The Unicode emojis in the tweets were converted to emoji short text with the help of a Python module called Emoji. The Python library Textblob was used to correct the spellings of many improperly spelled words. We use the NLTK library to remove stopwords and stem the corpus using Snowball Stemmer.

B. Data Annotation

We selected three graduate students[11] who are articulate in English to annotate the tweets with suitable complaint/non-complaint classes, severity levels, as well as emotion and sentiment tags. Before the annotation process began, the guidelines for annotation were provided.

[9]https://www.tweepy.org/
[10]https://developer.twitter.com/
[11]Annotators were recruited from the author's institution.

Table IV
Domain-wise data distribution in the *FINCORP* Dataset.

Domain	Non-Complaints	Complaints
Credit card	402	238
Customer service	12	697
Debit card	668	191
Economy	600	4
Financial policies	238	29
Investments	906	70
Loans	102	285
Retail banking	45	303
Salary	28	140
Transaction failure	132	1256
Total	3133	3149

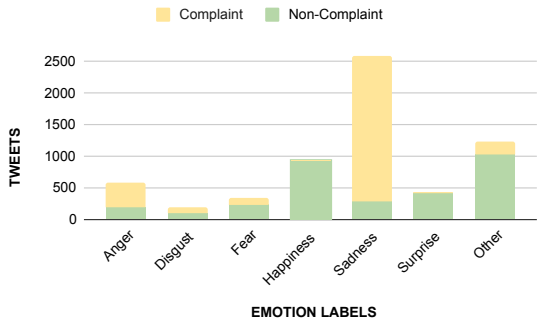

Figure 2. Distribution of emotion labels across *FINCORP* dataset.

Some sample cases outside the corpus were provided to train the annotators for the annotation work.
Annotators were selected using the following criteria:

- They are at least 22 years old and fluent in English (reading, writing, and speaking).
- Prepared to read, interpret, and annotate twitter instances according to the provided guidelines.

A few examples were provided before annotation for complaint severity level, and emotion task annotation as shown in Table II, III, respectively. Initially, we construct a binary annotation task to determine whether or not a tweet involves a complaint. Tweets are most often brief and convey a single idea. As a result, if a tweet includes a single complaint speech act, we consider the overall tweet to be a complaint. We utilize the complaint definition from earlier linguistic research [21] for complaint annotation: *A complaint presents a state of affairs that breaches the writer's clear expectation.*

We utilize the complaint severity levels (*no explicit reproach, disapproval, accusation, and blame*) defined in [2] as it is considered a benchmark in linguistic studies[12]. According to [2], classes are disconnected. In the 'No explicit reproach' class, the reason for the complaint is not specified, and the complaint is not harsh. The 'Disapproval' class conveys clear negative emotions, including discontent, anger, contempt, or disapproval. The primary distinction between 'Accusation' and 'Blame' is that the

[12]We assign a 'No Complaint Severity' class for non-complaint tweets.

Table V
Sample instances from the *FINCORP* dataset. Comp: Complaint, Non-Comp: Non-Complaint

Domain	Tweet	Label	Severity	Emotion	Sentiment
Loan	Why is your Panvel branch team not working on disbursing my loan which is approved since Friday?	Comp	Accusation	Sadness	Negative
Transaction Failure	I am not able to do online transaction from my credit card and customer care number also not responding.	Comp	Blame	Sadness	Negative
Credit Card	Rejecting Lost Credit Card Replacement without any valid reason! Using your card service for 10years. Highly disappointed with the way you treated your existing loyal customer.	Comp	Disapproval	Anger	Negative
Debit Card	I've found a Debit Card on my walk. I'd like to report it found. What do I do next?	Comp	No explicit reproach	Other	Neutral
Credit Card	Kpop concert tickets are great for my credit card points I got like $100 off a flight thanks.	Non-Comp	Non-Comp	Happiness	Positive

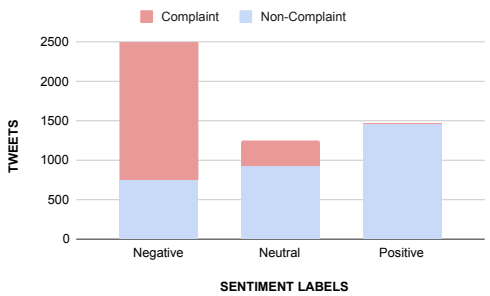

Figure 3. Distribution of sentiment polarities across *FINCORP* dataset

Table VI
Distribution of Complaint instances across the individual severity levels.

Severity class	Amount
Accusation	121
Blame	876
Disapproval	604
No explicit reproach	1548
Total	3149

complainant assumes the complainee is responsible for the violation in the latter. For the emotion annotation of the *FINCORP* dataset, we consider Ekman's [22] six basic emotions (*anger, disgust, fear, happiness, sadness, and surprise*). Other than these six basic emotions, we add a category 'Other' to represent such tweets that do not fall under the six emotion classes' scope. For the sentiment annotation, we consider three sentiment classes (*negative, neutral, positive*).
The final complaint, severity level, emotion, and sentiment labels are selected by utilizing the majority voting technique. The final annotated dataset excluded tweets with no common severity level, emotion, or sentiment label.

Finding agreement among the different annotators is vital in any annotation task requiring two or more annotators to obtain a credible annotated dataset. One such metric is Fleiss' kappa [23] score. We computed the Fleiss' kappa score on the annotated dataset to assess inter-rater agreement among the three annotators. We attained agreement scores of 0.83, 0.69, 0.68, and 0.82 on the complaint, severity level, emotion, and sentiment tasks, respectively, indicating that the annotations are of good quality.

C. Statistics

The *FINCORP* dataset comprises 6282 tweets with the corresponding complaint, severity, emotion, and sentiment labels. It consists of 3133 tweets in the non-complaint category (49.87%) and 3149 tweets in the complaint category (50.13%). Distributions of sentiment and emotion labels across the dataset are shown in Figure 2 and 3, respectively. Table IV presents the domain-wise data distribution statistics. Table V shows sample instances along with the corresponding complaint, severity, emotion, and sentiment labels. As can be observed from Figure 3, most of the complaint-type tweets have a negative polarity, with a few being neutral. In the case of emotion labels, most complaint tweets show sadness, and the next significant emotion is anger. The distribution of tweets across complaint severity classes is shown in Table VI.

IV. Methodology

In this section, we outline our problem and discuss the details of the baselines used for the evaluation of the dataset.

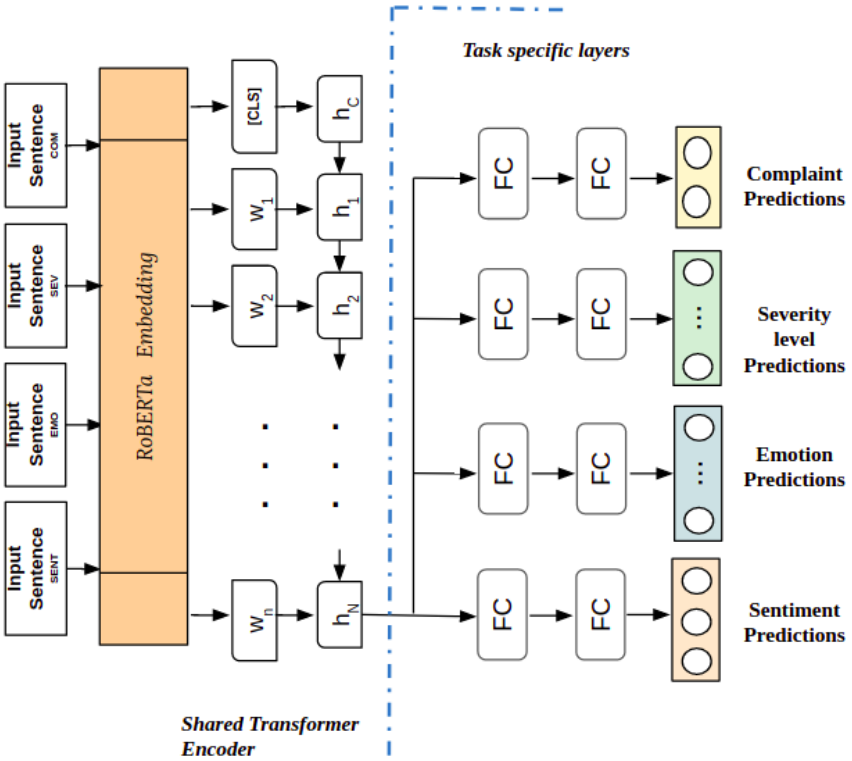

Figure 4. The architectural diagram of Multi-task Learning with RoBERTa (MTL-RoBERTa) framework. FC: Fully connected layers

A. Problem Definition:

We intend to learn four closely related tasks at the same time, including binary complaint identification (main-task), complaint severity prediction (main-task), emotion recognition, and sentiment analysis (auxiliary tasks). Let $(t_{\mathrm{i}}, e_{\mathrm{i}}, s_{\mathrm{i}}, m_{\mathrm{i}}, c_{\mathrm{i}})_{i=1}^{T}$ be a set of T instances where e_{i}, s_{i}, m_{i} and c_{i} represent the matching emotion, sentiment, severity and complaint labels for $t_{\mathrm{i}}{}^{\mathrm{th}}$ tweet, respectively. Here, $t_{\mathrm{i}} \epsilon T$, $e_{\mathrm{i}} \epsilon E$ (emotion classes), $s_{\mathrm{i}} \epsilon S$ (sentiment classes), $m_{\mathrm{i}} \epsilon M$ (severity classes) and $c_{\mathrm{i}} \epsilon C$ (binary complaint classes).

Our multi-task learning framework's objective is to maximize the function f (Eqn 1) that draws a new instance t_{i} to its fitting emotion label e_{i}, sentiment label s_{i}, severity label m_{i} and complaint label c_{i}, simultaneously.

$$argmax(\Pi_{k=0}^{R} P(m_i, s_i, e_i, c_i | t_i; \theta)), \tag{1}$$

where t_i is the input tweet whose complaint label (c_i), severity label (m_i), emotion label (e_i), and sentiment label (s_i) and are to be predicted. θ denotes the model's parameters we aim to optimize.

B. Multi-task learning with RoBERTa (MTL-RoBERTa):

Based on the work in [24], we employ multi-task learning with hard parameter sharing (MTL) to boost the binary complaint classification utilizing severity levels, emotion, and sentiment tasks. MTL with hard parameter sharing comprises a single encoder that is shared and modified by all the tasks and succeeded by task-specific branches. MTL allows two or more tasks to be learned simultaneously by sharing model knowledge and parameters. This boosts the binary complaint classification utilizing severity levels, emotion, and sentiment task features. The overall MTL-RoBERTa framework is shown in Figure 4.

Bidirectional Encoder Representations from Transformers (BERT) is a multi-layer bidirectional transformer encoder [25] based on the original work shown in [26]. It is trained on masked language modeling, which involves arbitrarily masking certain tokens from the input to predict them only based on context. RoBERTa [27] is an extension of BERT that has exhibited enhanced performance on social media analysis tasks [28], [29], by training on more data with varying hyperparameters.

The proposed MTL-RoBERTa model utilizes a single encoder layer that is shared and modified by all four tasks, with independent feed-forward networks for each task. We use the pre-trained RoBERTa model to embed the words of each tweet instance. The embedding representations are then fed to the shared transformer encoder (the 'roberta-base' model supports an inbuilt encoder as an attribute) to generate the instance's latent representation. This representation is then passed in parallel to four fully connected layers (for each of the four tasks), each of which has two hidden layers, to predict the output for each task separately. Each task-specific loss function is assigned a weight by the final loss function to be optimized. Finally, we add four separate output layers with softmax activation functions for binary and multi-class prediction.

Calculation of Loss: The categorical-cross entropy (CE) losses are calculated for the complaint (C), severity (M), emotion (E), and sentiment (S) tasks. Our proposed model's (MTL-RoBERTa) integrated loss function (L) is realized as follows:

$$L = a * L^{\mathrm{C}}{}_{\mathrm{CE}} + b * L^{\mathrm{M}}{}_{\mathrm{CE}} + c * L^{\mathrm{E}}{}_{\mathrm{CE}} + d * L^{\mathrm{S}}{}_{\mathrm{CE}}. \quad (2)$$

To calculate the overall loss, we calculated the weighted sum of the losses from the four tasks. The loss weights indicating the per-task loss-share to the overall loss are determined by the constants a, b, c, and d, which range from 0 to 1.

V. Experiments and Results

This section discusses the result and analysis of various baseline models and our proposed paradigm, as tested on the FINCORP corpus. The dataset and codes will be released publicly via the corresponding repository.

A. Baselines

Since *FINCORP* is a new dataset, for a comprehensive evaluation, we evaluate the dataset on multiple baselines, discussed as follows:

- **Logistic Regression (LR-BOW):** To measure the performance of the deep learning models, we also employ Logistic Regression, a common machine-learning-based classifier that has repeatedly served as a good baseline in many classification tasks. We employ Logistic Regression with standard Bag-of-Words (BOW) for CI and CS tasks.
- **Single-task systems:** For binary complaint identification and complaint severity prediction, we develop a single-task deep learning model based on RoBERTa ($\mathrm{STL_{CI}}$, $\mathrm{STL_{CS}}$).
- **Multi-task system:** Based on existing works in CI, we develop $\mathrm{Baseline_1}$[10] model as the multi-task baseline. $\mathrm{Baseline_1}$ is a shared private multi-task model for simultaneous classification of complaint, emotion, and sentiment tasks. For training the model, the authors employed three different sources of data. We also build an $\mathrm{MTL_{GloVe}}$ model wherein the textual embeddings are generated using the pre-trained GloVe[13] [30]. The word sequence encoder receives the embedding layer's output and analyses it to attain contextual knowledge from the text. The rest of the framework is the same.
- **Ablation models:** To evaluate the influence of complaint severity, emotion, and sentiment prediction on the complaint task independently and in varying combinations, we developed dual-task ($\mathrm{MTL_{CI+ER}}$, $\mathrm{MTL_{CI+SA}}$, $\mathrm{MTL_{CI+CS}}$) and tri-task variants ($\mathrm{MTL_{CI+ER+SA}}$, $\mathrm{MTL_{CI+ER+CS}}$, $\mathrm{MTL_{CI+SA+CS}}$) of our proposed model.

[13]GloVe:urlhttp://nlp.stanford.edu/data/wordvecs/glove.840B.300d.zip

B. Experimental Setup

We employ Python-based libraries to implement our proposed framework as well as the baselines, namely TensorFlow[14], Scikit-learn[15] [31]. The pre-trained model (RoBERTa) is from the open-source library[16] huggingface transformer. We ran all of the experiments on a single NVIDIA GeForce RTX 2080 Ti GPU. We evaluated the predictive performance of the proposed model and the baselines using accuracy and macro-F1 score. We have utilized 60% of the *FINCORP* dataset as training data, 15% for validation, and 25% as testing data for all the experiments. A seed value of 40 was used for a fair assessment of the models, allowing them to encounter the same training and testing data. To effectively employ the GPU, we kept the batch size at 8. We use *Adam* optimizer [32] with a learning rate of lr = 1e-3, $lr \epsilon$ {1e-1, 1e-2, 1e-3, 1e-4}. We use the base uncased model for RoBERTa, and fine-tune it with learning rate of lr = 5e-5, $lr \epsilon$ {1e-4, 5e-5, 1e-5, 1e-6}. The maximum sentence length is set at 50 characters, which spans 93% of the tweets in the training set. We utilize *ReLU* [33] activation and a *dropout* [34] of 30%, *dropout* ϵ {20%, 30%, 50%}. The output layers for complaint, severity, emotion, and sentiment classification tasks use $Softmax$ activation with 2, 4, 6, and 3 neurons, respectively. To train across all channels, categorical cross-entropy is employed as the loss function. *All of these parameter values were chosen following a thorough sensitivity study.*

C. Results

*1) **Results Discussion:**: Kindly note that the current paper's objective is to boost the efficacy of the CI and CS tasks by utilizing the two secondary tasks (ER and SA). As a result, we present the results and analysis with CI and CS as the primary tasks in all task pairings.*

The classification results from the various experiments are shown in Table VII. It can be observed that combining all associated tasks, such as severity, emotion, and sentiment, improves performance compared to single-task, dual-task, and tri-task variants. This improvement indicates that the proposed model makes effectual usage of interaction between the four tasks. As seen in Table VII, the MTL-RoBERTa model, which includes all four tasks (CI, CS, ER, and SA), beats the single-task complaint and severity variants ($\mathrm{STL_{CI}}$, $\mathrm{STL_{CS}}$). $\mathrm{MTL_{CI+CS}}$ outperforms $\mathrm{MTL_{CI+ER}}$ and $\mathrm{MTL_{CI+SA}}$ in the dual-task variants. This supports our assumption that the complaint severity level information can boost the complaint identification task while simultaneously learning these two tasks. This could also be influenced by the notion that sentiment by itself is usually unable to convey essential details about the complainant's predicament. For instance, negative feelings toward a company or a product

[14]https://www.tensorflow.org/
[15]https://scikit-learn.org/stable/
[16]https://huggingface.co/model and https://tfhub.dev/google/collections/bert/1

Table VII
Results of the proposed model and other baselines with respect to macro-F1 score and accuracy value for CI task. Bold-faced values indicate the maximum scores achieved. The * indicates that the results are statistically significant.

Model	Accuracy (%)	F1-score(%)
$SOTA_{CI}$[16]	84.26 ± .15	83.53 ± .03
LR-BOW	67.52 ± .03	64.28 ± .03
$Baseline_1$[10]	82.38 ± 1.02	81.84 ± 1.05
MTL_{GloVe}	80.45 ± 1.03	79.32 ± 1.02
Single-task baseline		
STL_{CI}	81.26 ± 0.03	80.16 ± 0.05
Dual-task baselines		
MTL_{CI+ER}	81.79 ± 1.07	81.54 ± 1.13
MTL_{CI+SA}	80.58 ± 1.12	79.51 ± 1.06
MTL_{CI+CS}	83.87 ± 1.04	83.28 ± 1.07
Tri-task baselines		
$MTL_{CI+ER+SA}$	82.48 ± 1.17	82.14 ± 1.03
$MTL_{CI+ER+CS}$	84.28 ± 1.05	83.54 ± 1.02
$MTL_{CI+SA+CS}$	83.39 ± 1.02	81.89 ± 1.06
Quad-task baseline		
MTL-RoBERTa	**86.33*** ± .03	**85.82*** ± .05

Table VIII
Results of all the proposed model and other baselines with respect to the macro-F1 score and accuracy value for CS task. Bold-faced values indicate the maximum scores achieved. The * indicates that the results are statistically significant.

Model	Accuracy (%)	F1-score(%)
$SOTA_{CS}$[15]	65.67 ± .11	64.17 ± .12
LR-BOW	48.52 ± .14	46.28 ± .12
MTL_{GloVe}	60.49 ± 1.03	60.12 ± 0.14
Single-task baseline		
STL_{CS}	57.89 ± 1.04	57.19 ± 1.12
Dual-task baselines		
MTL_{CS+CI}	61.22 ± 1.14	60.08 ± 1.03
MTL_{CS+ER}	58.67 ± 1.02	56.88 ± 1.04
MTL_{CS+SA}	57.88 ± 1.11	57.37 ± 1.03
Tri-task baselines		
$MTL_{CS+ER+SA}$	62.58 ± 1.07	61.74 ± 1.03
$MTL_{CS+ER+CI}$	64.68 ± 1.07	63.14 ± 1.13
$MTL_{CS+SA+CI}$	63.69 ± 1.12	60.19 ± 1.04
Quad-task baseline		
MTL-RoBERTa	**67.89*** ± .15	**66.52*** ± .12

might be influenced by numerous emotions such as anger, sorrow, etc. Consequently, distinct or minor variations in emotion cannot be accurately determined and conveyed solely through sentiment.

Specifically, in the case of tri-task variants, $MTL_{CI+ER+CS}$ outperforms other tri-task baselines with the macro-F1 score of 83.54%. One of the reasons for the inferior results of $MTL_{CI+ER+SA}$ and $MTL_{CI+SA+CS}$ variants could be that the sentiment analysis task does not contribute much to the multi-task architecture as compared to severity and emotion tasks.

In Table VIII, we report the results of the MTL-RoBERTa model for the complaint severity task. Overall, the MTL-RoBERTa model yields the best result with the macro-F1 score of 66.52%. Moreover, of all the baselines, $MTL_{CS+ER+CI}$ outperforms other baselines with the macro-F1 score of 63.14%. This is in line with [2], where the author claims that the way people convey their grievances is linked to their emotional states.
The findings reported here are all statistically significant.[17] *[35]..*

2) ***Comparison with State-of-the-art (SOTA)::*** For single-task CI, we compare the MTL-RoBERTa model to the existing state-of-the-art technique [16]. $SOTA_{CI}$ makes use of a variety of neural language models that are aided by transformer networks. We also compare the results of the MTL-RoBERTa model with those of the existing state-of-the-art approach [15] for single-task CS task. $SOTA_{CS}$ uses a wide range of neural language models with several linguistic features augmented by transformer networks. We re-implement it on the FINCORP dataset and report the results in Table VII and VIII. The MTL-RoBERTa model outperforms the SOTA techniques for complaint identification and severity prediction tasks.

3) ***Qualitative Analysis::*** Table IX presents a few examples of MTL-RoBERTa and single-task (complaint task) model predictions. Even though the 1st and 2nd cases in Table IX communicate dissatisfaction in a less explicit tone, the MTL-RoBERTa model accurately predicts the examples as complaints. Even though the 3rd and 4th examples are indirect and vague in nature, the MTL_{All} model identifies that there is no violation of trust. These examples highlight the importance of detecting complaints based on severity, emotion, and sentiment knowledge.

4) ***Error Analysis::*** We examine some of the plausible reasons for the MTL-RoBERTa model's complaint and severity prediction inaccuracies:

- **Elusive Complaints:** Non-complaints are incorrectly predicted when complaints are communicated in an implicit or cryptic manner. One justification for this could be the complainant's lack of explicit accusation or blame. For example, *'Why am I waiting 60mins just so I can ok a debit card transaction!?'*. The predicted class is non-complaint, whereas the instance's actual class is complaint. Unless the tweet does not mention the particular company or use a critical phrase, the model misinterprets it as non-complaint.
- **Neighbouring Severity Level Errors:** Instances that belong to neighboring severity levels tend to be semantically and compositionally similar, sometimes leading to misclassification. For example, *'You are looting consumer's money. It's very disappointing behavior that they didn't refund back failed transaction money even after weeks and their costumer support show us the way to our bank, who doesn't recognize their transaction number.'*; predicted severity level: 'disapproval'. The correct severity level for the preceding example is 'accusation', but due to the presence of negative terms such as *disappointing* which

[17]For the significance test, we utilized the Student's t-test. When testing the null hypothesis, the results are statistically significant (p-value < 0.05).

Table IX
Few example instances along-with associated predicted labels for the best performing multi-task (MTL-RoBERTa) model and its single-task CI equivalent. (STL_{CI}).

Sentence	MTL-RoBERTa	STL_{CI}	Actual Label
This has been a great way to spend an afternoon, trying to "chase" down that $832 deposit-student loan funds-that disappeared, then you mysteriously closed the account	Complaint	Non-Complaint	Complaint
I had to report my visa as STOLEN after purchasing from Amazon! Thanks alot. And no proper response from the team.	Complaint	Non-Complaint	Complaint
I actually don't prefer AMEX. A great card but I only use cards that have a zero annual fee. I personally think annual fees on a credit card are ridiculous.	Non-Complaint	Complaint	Non-Complaint
Great job, dude why there's no crypto payment for VIP membership?	Non-Complaint	Complaint	Non-Complaint

usually appear in the 'disapproval' severity class, the proposed model misclassifies it.

- **Satirical Instances:** Instances where the underlying tone is positive or neutral, but the instance is of complaint type, the MTL-RoBERTa model inaccurately predicts such instances as non-complaint. For example, *'Yes, I need to pay a transaction that cannot be done online/over the phone because it exceeds the daily limit.'*. For this instance, the correct class is complaint, but the model predicts it as non-complaint. One reason could be the neutral undertone and the usage of less explicit words to signify complaint.
- **Probing Complaints:** We also identified that the model incorrectly labels cases of weak discontent with a inquiring tone as non-complaints. For example, *'What is this restaurant packaging charges and where is this money going when McDonald's on their website is not charging it?'*. The correct class for the given example is complaint. The model classifies it as non-complaint because there is no clear objection, and the sentence is interrogative. To effectively classify such occurrences, the system requires background information and knowledge about the key aspects of the sentence.

VI. Conclusion

In this work, we have introduced FINCORP, an annotated dataset of complaints originating between financial institutions and consumers on Twitter. We have addressed the task of complaint identification and severity level prediction (primary tasks) jointly using a multi-task framework assisted by emotion and sentiment detection as auxiliary tasks. The paper goes into a detailed description of the dataset and the complete process of creating it. Every instance in the FINCORP dataset has been annotated across four axes: complaint, severity level, emotion, and sentiment classes, resulting in a dataset with a lot of diversity. We evaluated the dataset and reported the findings using several single-task, multi-task and other existing baselines. Given the scarcity of task-specific (complaint identification) data in English and even some low-resource languages, we believe this dataset would add value to social media analytics research and practice.

We plan to expand our work in complaint identification at the sentence level in english as well as code-mixed languages in the future. Furthermore, the annotated corpus can aid research in other emotion and sentiment classifications tasks.

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